CAT 2024 Logical Reasoning: Linear & Vertical Arrangements Master Solved Guide

Solved MCQ Handbook & Concept Master Guide

Executive Concept & Formula Summary

- **Linear Horizontal Arrangement**:
- **Facing North**: Your Left = Person's Left; Your Right = Person's Right (you sit/stand behind them).
- **Facing South**: Your Left = Person's Right; Your Right = Person's Left (they face towards you).
- **Default Assumption**: If no direction is mentioned, assume everyone is facing **North**.
- **Extreme Ends**: Linear arrangements have fixed boundary points (Left Extreme and Right Extreme), unlike circular arrangements.
- **"Right/Left" vs. "Immediate Right/Left"**:
- *"A is to the right of B"* implies A can occupy **any** seat to the right of B.
- *"A is to the immediate right of B"* means A sits directly adjacent to B on B's right side.
- **Vertical Arrangement (Floors / Queues / Stacks)**:
- **Floor Numbering Rules**: In standard 7-floor problems without a ground floor, floors are indexed sequentially from 1 (bottom-most) to 7 (top-most).
- **Floors Above / Below**:
- *"X is N floors above Y"* means **Floor**(X) = **Floor**(Y) + N.
- *"X lives right/immediately above Y"* means **Floor**(X) = **Floor**(Y) + 1.
- **Center Position**: For N items where N is odd, the exact middle position is (N+1)/2.

Q#	Topic / Concept Tested	Correct Answer	Core Key Takeaway
Q1	Horizontal Arrangement - Seat Indexing	**(C)**	Fill definite clues first; seat 7 is the right extreme.
Q2	Horizontal Arrangement - Relative Neighbor Analysis	**(C)**	"Left" and "Right" must align with the North-facing perspective.
Q3	Horizontal Arrangement - Middle Seat Identification	**(B)**	For 7 positions, seat (7+1)/2 = 4 is the exact middle seat.
Q4	Horizontal Arrangement - Reversed Seat Numbering	**(A)**	Re-numbering from the right end shifts Seat k to position (8-k).
Q5	Horizontal Arrangement - Intermediary Count	**(B)**	Count only the seats strictly lying between two given positions.
Q6	Vertical Arrangement - Floor Location Logic	**(D)**	Top floor resident (Floor 7) has no resident living above her.
Q7	Vertical Arrangement - Symmetric Middle Floor	**(C)**	The 4th floor has exactly 3 floors above (5,6,7) and 3 below (1,2,3).
Q8	Vertical Arrangement - Prime Sum Calculation	**(C)**	Sum of floor numbers: Ryan (2) + Roy (3) = 5 (Prime).
Q9	Vertical Arrangement - Relative Floor Difference Equation	**(A)**	Equate differences: **Floor**(**Ram**) - **Floor**(**Host**) = **Floor**(**Rwanda**) - **Floor**(**Ryan**).
Q10	Vertical Arrangement - Neighbor Elimination Technique	**(D)**	Directly identify adjacent floor pairs from the problem statements.

Question 1: Horizontal Arrangement - Identifying End Positions

***Topic**: Logical Reasoning > Linear Horizontal Arrangement | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Easy*

Q. Seven students - Peter, Quill, Raul, Sabir, Tim, Urvashi, and Vimal - sit on a bench facing the stage towards the North. The arrangement follows these clues:

1. Urvashi is to the immediate right of Tim.
2. Tim is 4th to the right of Vimal.
3. The student seated 3rd to the left of Sabir is seated at one of the extreme ends.
4. Raul is a neighbor of both Quill and Sabir.

If the seats are numbered 1 to 7 from the left end, in which seat is Peter seated?

(A) 4

(B) 6

(C) 7

(D) 5

CORRECT ANSWER: (C)

Core Reason: By fixing Sabir at seat 4, Raul at seat 3, Quill at seat 2, and Vimal at seat 1, Tim sits at seat 5 and Urvashi at seat 6, leaving seat 7 for Peter.

Step-by-Step Explanation

1. **Identify Direction & Slots**: Draw 7 slots numbered 1 to 7 from left to right. Since everyone faces North, Left = Our Left and Right = Our Right.
2. **Apply Direct Clues**:

- Clue 3: Someone sitting 3rd to the left of Sabir is at an extreme. If Sabir were placed such that his left reaches the right extreme, he would fall off the bench. Thus, the student 3rd to his left must be at Seat 1 (the Left Extreme).
- Counting 3 seats right from Seat 1: Seat 1 -> Seat 2 (1st right) -> Seat 3 (2nd right) -> Seat 4 (3rd right). Thus, **Sabir is at Seat 4**.

3. **Analyze Clue 4**: Raul is adjacent to both Quill and Sabir.

- *Case 1*: Quill = Seat 2, Raul = Seat 3, Sabir = Seat 4.
- *Case 2*: Sabir = Seat 4, Raul = Seat 5, Quill = Seat 6.

4. **Test Clue 2 & Clue 1**: Tim is 4th to the right of Vimal.

- In *Case 1*: Place Vimal at Seat 1. 4th to the right of Seat 1 is Seat 1 + 4 = 5 (Tim).
- Then by Clue 1, Urvashi is to the immediate right of Tim -> Seat 6.
- Remaining seat is Seat 7, occupied by **Peter**.
- In *Case 2*: If Vimal sits at Seat 3, Tim sits at Seat 7. But Urvashi must be to Tim's immediate right, which would put her outside Seat 7 (invalid). Thus Case 2 is eliminated.

5. **Final Seating Arrangement (Seats 1 to 7)**:

Seat 1: Vimal, Seat 2: Quill, Seat 3: Raul, Seat 4: Sabir, Seat 5: Tim, Seat 6: Urvashi, Seat 7: Peter

Option Analysis

- **Option (A) [INCORRECT]**: Seat 4 is occupied by Sabir.
- **Option (B) [INCORRECT]**: Seat 6 is occupied by Urvashi.
- **Option (C) [CORRECT]**: Seat 7 is occupied by Peter.
- **Option (D) [INCORRECT]**: Seat 5 is occupied by Tim.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

Always test boundary constraints! If a person is on seat 7 and someone sits to their "immediate right", that case is invalid because seat 8 does not exist.

Question 2: Relative Positions of Immediate Neighbors

Topic: Logical Reasoning > Linear Horizontal Arrangement | Exam / Year: CAT 2024 / LRDI Starter Kit | Difficulty: Easy

Q. Based on the bench seating arrangement of the seven students (Vimal, Quill, Raul, Sabir, Tim, Urvashi, Peter) facing North, fill in the blanks: `______ sits to the left and ______ sits to the right of Quill.`

(A) Tim sits to the left and Raul sits to the right

(B) Raul sits to the left and Vimal sits to the right

(C) Vimal sits to the left and Raul sits to the right

(D) Urvashi sits to the left and Sabir sits to the right

CORRECT ANSWER: (C)

Core Reason: Quill is at Seat 2. Vimal is at Seat 1 (to Quill's left) and Raul is at Seat 3 (to Quill's right).

Step-by-Step Explanation

1. From the seating arrangement established in Question 1:

[1: Vimal] [2: Quill] [3: Raul] [4: Sabir] [5: Tim] [6: Urvashi] [7: Peter]

2. Locate Quill's position: **Seat 2**.
3. Identify the person to Quill's left: **Vimal** (Seat 1).
4. Identify the person to Quill's right: **Raul** (Seat 3).
5. Therefore, Vimal sits to the left and Raul sits to the right of Quill.

Option Analysis

- **Option (A) [INCORRECT]**: Tim is at Seat 5, not to the immediate left of Quill.
- **Option (B) [INCORRECT]**: Reverses the order of left and right neighbors.
- **Option (C) [CORRECT]**: Accurately reflects Vimal on the left (Seat 1) and Raul on the right (Seat 3).
- **Option (D) [INCORRECT]**: Urvashi is at Seat 6, far away from Quill.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

Pay strict attention to the order specified in the blank: "Left" comes first, "Right" comes second. Option (B) swaps the order as a classic distractor!

Question 3: Identifying the Central Position

Topic: Logical Reasoning > Linear Horizontal Arrangement | Exam / Year: CAT 2024 / LRDI Starter Kit | Difficulty: Easy

Q. Who sits in the exact middle of the bench?

(A) Raul

(B) Sabir

(C) Tim

(D) More than one of the above

CORRECT ANSWER: (B)

Core Reason: For 7 linearly arranged seats, the 4th seat is the middle position, which is occupied by Sabir.

Step-by-Step Explanation

1. The total number of seats is N = 7.
2. The exact middle seat formula for an odd number of positions is:

Middle Seat = (N + 1)/2 = (7 + 1)/2 = 4

3. Referring to the seating arrangement:

Seat 4 = **Sabir**

4. Hence, Sabir sits in the middle with exactly 3 people to his left (Vimal, Quill, Raul) and 3 people to his right (Tim, Urvashi, Peter).

Option Analysis

- **Option (A) [INCORRECT]**: Raul sits at Seat 3.
- **Option (B) [CORRECT]**: Sabir sits at Seat 4, which is the exact center.
- **Option (C) [INCORRECT]**: Tim sits at Seat 5.
- **Option (D) [INCORRECT]**: There is only one unique middle seat when N is odd.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

If N were an even number (e.g., 8 seats), there would be two middle seats (Seats 4 and 5), in which case options like "More than one" or "Cannot be uniquely determined" could apply.

Question 4: Reversed Indexing and Summation

***Topic**: Logical Reasoning > Linear Horizontal Arrangement | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Moderate*

Q. If the seats are re-numbered from 1 to 7 starting from the right end, what is the sum of the new seat numbers of Quill, Tim, and Urvashi?

(A) 11

(B) 10

(C) 12

(D) 9

CORRECT ANSWER: (A)

Core Reason: Numbering from the right end gives Urvashi = 2, Tim = 3, and Quill = 6. Sum = 2 + 3 + 6 = 11.

Step-by-Step Explanation

1. List the students in order from Left to Right:

Vimal (1), Quill (2), Raul (3), Sabir (4), Tim (5), Urvashi (6), Peter (7)

2. Re-number the seats from the **Right End** (1' to 7'):

- Peter = Seat 1'
- Urvashi = Seat 2'
- Tim = Seat 3'
- Sabir = Seat 4'
- Raul = Seat 5'
- Quill = Seat 6'
- Vimal = Seat 7'

3. Extract the requested new seat numbers:

- Quill = 6
- Tim = 3
- Urvashi = 2

4. Calculate the required sum:

Sum = 6 + 3 + 2 = 11

Option Analysis

- **Option (A) [CORRECT]**: Sum calculation 6 + 3 + 2 = 11 is mathematically exact.
- **Option (B) [INCORRECT]**: Arithmetic miscalculation.
- **Option (C) [INCORRECT]**: Miscounting positions by using the old left-to-right indices.
- **Option (D) [INCORRECT]**: Off-by-one error during re-indexing.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

Conversion Rule: For N items, if a position is k from the left, its position from the right is (N + 1 - k).

- Quill: 8 - 2 = 6
- Tim: 8 - 5 = 3
- Urvashi: 8 - 6 = 2

Question 5: Intermediary Person Count

***Topic**: Logical Reasoning > Linear Horizontal Arrangement | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Easy*

Q. How many people are seated between Sabir and Peter?

(A) 1

(B) 2

(C) 3

(D) 0

CORRECT ANSWER: (B)

Core Reason: Sabir is at Seat 4 and Peter is at Seat 7. The seats strictly between them are Seats 5 and 6 (Tim and Urvashi), giving a count of 2.

Step-by-Step Explanation

1. Identify the seat numbers of the two persons:

Seat of Sabir = 4, **Seat of Peter** = 7

2. Identify the seats strictly between 4 and 7:

Intermediary Seats = {5, 6}

3. Count the occupants of these seats:

- Seat 5: Tim
- Seat 6: Urvashi

4. Total count of people seated between them = 2.

Option Analysis

- **Option (A) [INCORRECT]**: Counts only one person.
- **Option (B) [CORRECT]**: Correctly identifies 2 individuals (Tim and Urvashi).
- **Option (C) [INCORRECT]**: Includes Sabir or Peter in the count.
- **Option (D) [INCORRECT]**: Mistakenly assumes they sit adjacent to each other.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

"Between X and Y" excludes X and Y themselves. Use the formula: **Count** = |**Pos**(Y) - **Pos**(X)| - 1 = |7 - 4| - 1 = 2.

Question 6: Vertical Floor Arrangement - Top Floor Location

***Topic**: Logical Reasoning > Vertical Arrangement (Floors) | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Easy*

Q. Seven residents - Ram, Reet, Rishi, Roy, Rutvi, Ryan, and Rwanda - live on seven different floors of an apartment building numbered 1 to 7 (with no ground floor). The following clues are provided:

1. Ram lives right above Rwanda.
2. Rwanda's house is 4 floors above Rishi.
3. Roy lives between Reet and Ryan.
4. Rishi lives 3 floors below Reet.
5. Rishi lives either on the 1st or the 7th floor.

Where does Rutvi live?

(A) Immediately below Ram

(B) Right above Ryan

(C) Between Ryan and Rwanda

(D) None of the above

CORRECT ANSWER: (D)

Core Reason: Rutvi lives on the 7th floor (the topmost floor). None of the options correctly describe her position relative to others.

Step-by-Step Explanation

1. **Floor Setup**: 7 floors, numbered 1 (bottom) to 7 (top).
2. **Apply Clue 5 & Clue 4**:

- Rishi lives on Floor 1 or Floor 7.
- Clue 4: "Rishi lives 3 floors below Reet."
- If Rishi were on Floor 7, Reet would have to be on Floor 7 + 3 = 10 (impossible).
- Therefore, **Rishi lives on Floor 1**.
- Reet lives 3 floors above Rishi -> Floor 1 + 3 = 4. (**Reet = Floor 4**).

3. **Apply Clue 2**:

- Rwanda's house is 4 floors above Rishi -> Floor 1 + 4 = 5. (**Rwanda = Floor 5**).

4. **Apply Clue 1**:

- Ram lives right above Rwanda -> Floor 5 + 1 = 6. (**Ram = Floor 6**).

5. **Apply Clue 3**:

- Roy lives between Reet (Floor 4) and Ryan.
- Since Floor 5 is occupied by Rwanda, Ryan must live on Floor 2, putting **Roy on Floor 3**.

6. **Assign Remaining Person**:

- Floors 1 to 6 are occupied by Rishi (1), Ryan (2), Roy (3), Reet (4), Rwanda (5), Ram (6).
- Floor 7 remains, which must be occupied by **Rutvi**.

7. **Evaluate Options for Rutvi (Floor 7)**:

- Below Ram? No, she is above Ram.
- Right above Ryan? No, she is above Ram.
- Between Ryan and Rwanda? No.

- Thus, **None of the above** is correct.

Option Analysis

- **Option (A) [INCORRECT]**: Rutvi is above Ram (Floor 6), not below.
- **Option (B) [INCORRECT]**: Roy is right above Ryan (Floor 2), not Rutvi.
- **Option (C) [INCORRECT]**: Roy is between Ryan and Reet.
- **Option (D) [CORRECT]**: Rutvi lives on the 7th floor, which is not correctly described by options A, B, or C.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

"N floors above X" means **Floor**(X) + N, NOT **Floor**(X) + N + 1. 3 floors below Floor 4 is 4 - 3 = 1.

Question 7: Identifying Symmetric Floor Positions

***Topic**: Logical Reasoning > Vertical Arrangement (Floors) | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Easy*

Q. Who among the following has an equal number of floors above and below their floor?

(A) Roy

(B) Ram

(C) Reet

(D) Rwanda

CORRECT ANSWER: (C)

Core Reason: Reet lives on Floor 4, which has exactly 3 floors above it (5, 6, 7) and 3 floors below it (1, 2, 3).

Step-by-Step Explanation

1. From the solved building arrangement:

Floor 7: Rutvi, Floor 6: Ram, Floor 5: Rwanda, Floor 4: Reet, Floor 3: Roy, Floor 2: Ryan, Floor 1: Rishi

2. Count the number of floors above and below each resident:

- **Floor 4 (Reet)**:

- Floors above: 5, 6, 7 (3 floors)

- Floors below: 1, 2, 3 (3 floors)

3. Equal number of floors above and below indicates the person lives on the central floor (**Floor 4**).

4. Therefore, **Reet** satisfies the condition.

Option Analysis

- **Option (A) [INCORRECT]**: Roy lives on Floor 3 (4 floors above, 2 below).

- **Option (B) [INCORRECT]**: Ram lives on Floor 6 (1 floor above, 5 below).

- **Option (C) [CORRECT]**: Reet lives on Floor 4 (3 floors above, 3 below).

- **Option (D) [INCORRECT]**: Rwanda lives on Floor 5 (2 floors above, 4 below).

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

The middle floor of an N-floor building (where N is odd) is always (N+1)/2. For N=7, the middle floor is (7+1)/2 = 4.

Question 8: Number Theory Logic in Floor Problems

***Topic**: Logical Reasoning > Vertical Arrangement / Quant Reasoning | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Moderate*

Q. Which of the following pairs of residents live on floors whose floor numbers sum up to a prime number?

(A) Rishi and Roy

(B) Ram and Ryan

(C) Ryan and Roy

(D) Rutvi and Rwanda

CORRECT ANSWER: (C)

Core Reason: Ryan lives on Floor 2 and Roy lives on Floor 3. The sum of their floor numbers is 2 + 3 = 5, which is a prime number.

Step-by-Step Explanation

1. List all residents with their corresponding floor numbers:

- Rishi = 1
- Ryan = 2
- Roy = 3
- Reet = 4
- Rwanda = 5
- Ram = 6
- Rutvi = 7

2. Test each option pair:

- **Option (A)**: Rishi (1) + Roy (3) = 1 + 3 = 4 (Composite number).
- **Option (B)**: Ram (6) + Ryan (2) = 6 + 2 = 8 (Composite number).
- **Option (C)**: Ryan (2) + Roy (3) = 2 + 3 = 5 (Prime number).
- **Option (D)**: Rutvi (7) + Rwanda (5) = 7 + 5 = 12 (Composite number).

3. Hence, Option (C) gives a prime number sum.

Option Analysis

- **Option (A) [INCORRECT]**: Sum = 4, which is divisible by 2.
- **Option (B) [INCORRECT]**: Sum = 8, which is divisible by 2 and 4.
- **Option (C) [CORRECT]**: Sum = 5, which has no divisors other than 1 and 5.
- **Option (D) [INCORRECT]**: Sum = 12, which is divisible by 2, 3, 4, 6.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

Remember that 2, 3, 5, 7, 11, 13, 17, 19... are prime numbers. 1 is neither prime nor composite!

Question 9: Relative Floor Difference Calculations

***Topic**: Logical Reasoning > Vertical Arrangement (Algebraic Clues) | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Moderate*

Q. Raghu, a guest visiting the building, goes to a floor that is as far below Ram's floor as Ryan's floor is below Rwanda's floor. Whose guest is Raghu?

(A) Roy

(B) Reet

(C) Rishi

(D) Ryan

CORRECT ANSWER: (A)

Core Reason: Ryan (Floor 2) is 3 floors below Rwanda (Floor 5). Counting 3 floors below Ram (Floor 6) gives Floor 3, which belongs to Roy.

Step-by-Step Explanation

1. Identify the floor numbers of the relevant persons:

- Ram = Floor 6
- Rwanda = Floor 5
- Ryan = Floor 2

2. Calculate the floor difference between Rwanda and Ryan:

Delta = **Floor**(**Rwanda**) - **Floor**(**Ryan**) = 5 - 2 = 3 **floors**

3. Set up the equation for the floor visited by Raghu ($\textbf{Floor}_X$):

$\textbf{Floor}(\textbf{Ram}) - \textbf{Floor}_X$ = Delta $6 - \textbf{Floor}_X = 3$ implies $\textbf{Floor}_X = 6 - 3 = 3$

4. Check who lives on Floor 3: **Roy lives on Floor 3**.

5. Therefore, Raghu is visiting Roy.

Option Analysis

- **Option (A) [CORRECT]**: Floor 3 belongs to Roy, matching the required difference equation.
- **Option (B) [INCORRECT]**: Reet lives on Floor 4 (6 - 4 = 2, incorrect gap).
- **Option (C) [INCORRECT]**: Rishi lives on Floor 1.
- **Option (D) [INCORRECT]**: Ryan lives on Floor 2.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

Convert verbal relational statements directly into algebraic equations: **Ram** - X = **Rwanda** - **Ryan**. Avoid guessing visually!

Question 10: Neighbor Elimination from Statement Clues

***Topic**: Logical Reasoning > Arrangement Clue Elimination | **Exam / Year**: CAT 2024 / LRDI Starter Kit | **Difficulty**: Easy*

Q. Based on the floor arrangement, which of the following pairs are NOT immediate neighbors?

(A) Roy and Reet

(B) Ram and Rwanda

(C) Ryan and Roy

(D) Rishi and Reet

CORRECT ANSWER: (D)

Core Reason: Rishi lives on Floor 1 and Reet lives on Floor 4. They are separated by two floors and are not immediate neighbors.

Step-by-Step Explanation

1. Examine the floor levels of the given pairs:

- **Roy and Reet**: Floor 3 and Floor 4 -> Immediate neighbors (|4 - 3| = 1).
- **Ram and Rwanda**: Floor 6 and Floor 5 -> Immediate neighbors (|6 - 5| = 1).
- **Ryan and Roy**: Floor 2 and Floor 3 -> Immediate neighbors (|3 - 2| = 1).
- **Rishi and Reet**: Floor 1 and Floor 4 -> Not neighbors (|4 - 1| = 3).

2. The question asks for the pair that are **NOT** neighbors.
3. Therefore, Option (D) is the correct answer.

Option Analysis

- **Option (A) [INCORRECT]**: Roy (3) and Reet (4) are adjacent.
- **Option (B) [INCORRECT]**: Ram (6) and Rwanda (5) are adjacent.
- **Option (C) [INCORRECT]**: Ryan (2) and Roy (3) are adjacent.
- **Option (D) [CORRECT]**: Rishi (1) and Reet (4) have a gap of 2 floors between them, so they are NOT neighbors.

EXAM TIP / MNEMONIC - Exam Trap / Key Memory Rule

Freebie Alert: In competitive exams like CAT/UPSC, questions asking "which pair is NOT a neighbor" can often be answered by simply reading the clues without even drawing the entire diagram, as the initial clues state that Roy is between Reet and Ryan, and Ram is right above Rwanda!